

Simple, Low Frequency, Fast Recovery Crossed Coil Probe for Pulsed NMR

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Abstract. The realization of a crossed coil probe for pulsed NMR showing a recovery time of 5 μ s at 10 MHz is discussed. The system, which is based on a transitory damping of the transmitter coil and on the simultaneous grounding of the receiver coil, does not need a particular isolation between the two sections.

The possibility of performing an experiment of NMR in samples with a rapid transverse relaxation, as it happens in solids, is tied up to the necessity of getting a short dead time in the spectrometer after the transmitter pulse, so that the FID could be detected without appreciable distortion. This time interval is usually called the recovery time of the system and generally arises from an electronic recovery time and a ringing down time. The former is dependent on the characteristics of the receiver [1], while the latter is determined by the fact that the transmitter signal, stored by the tuned probe, decays to the noise level from the initial amplitude V_0 according to the relationship [2,3]:

$$V = V_0 \exp(-t/\tau), \quad (1)$$

where $\tau = 2Q/\omega$ is the ringing down constant.

Eq.(1) shows that, at a fixed quality factor (Q) value, the ringing down of the probe may be crucial at low frequencies. On the other hand, an excess lowering of Q should be avoided since the nuclear signal-to-noise (S/N) ratio scales as $Q^{1/2}$. Actually, if one assumes a value of 50 for the factor Q , Eq.(1) provides a ringing down time of 32 μ s for a RF pulse amplitude of some kV at a frequency of 10 MHz.

In single coil configurations, the widespread use of crossed diodes and quarter wave lines between the probe and the receiver, as dictated by isolation and matching requirements [4], is not sufficient to prevent electronic saturation and does not result in any significant shortening of the ringing down time. This problem has been so far approached mainly with two kinds of methods: passive limiting and active gating.

Passive methods have regarded, in some cases [5,6], the limitation of the RF level to the receiver; however the resolution time so far achieved was limited by the transmitter coil ring down. Most of these passive methods, however, involve either a permanent [2,7–9] or a temporary [10–13] damping of the Q , with varying degrees of success. While a permanent damping implies an obvious degradation of S/N ratio, a temporary lowering of the quality factor may lead to some drawbacks (as anew ringing or extra noise), as widely discussed by Hoult [14].

The best results are obtained with active methods [3,14], consisting in a unitary approach to the probe-receiver system, in which the probe is generally damped in its high voltage part of the ringing down; the low voltage part being shortened with the aid of a negative feedback driving the preamplifier stage of the receiver. However these techniques are rather critical and quite involved in their practical realization.

The crossed coil configuration could represent a suitable arrangement to minimize the recovery time, provided that a very high isolation between the transmitter and the receiver coil can be achieved. Actually, a nominal isolation value of approximately 120 dB is required [1]; such a value being however extremely difficult to reach from both mechanical and electrical point of view [15,16]. Actually, even if a particular attention is paid both to the geometrical and electrical isolation, the maximal value which can be obtained is around 70 dB [1,17], that is very far from the optimal value.

In the present work we discuss the realization of a crossed coil probe, especially designed for our NMR experiments in solids, which couples its ability to reduce the recovery time down to 5 μ s at 10 MHz to a remarkable simplicity in its realization.

It should be remarked that this crossed probe, whose transmitting section is damped after the RF pulse, while its detection part is held grounded during this pulse, does not necessarily need a high isolation between the two sections. Thus, such use of a crossed coil configuration if, on one hand, it does not involve a tedious and difficult mismatching, it offers, on the other, several advantages with respect to the single coil. Examples of those are the possibility of longitudinal detection [18], an higher B_1 homogeneity and the possibility of matching independently transmitter and receiver parts. In particular, tuning the transmitter coil at low impedance enables to work with lower voltage, avoiding the problems connected with tuning capacitors breakdown and expensive RF power supplies.

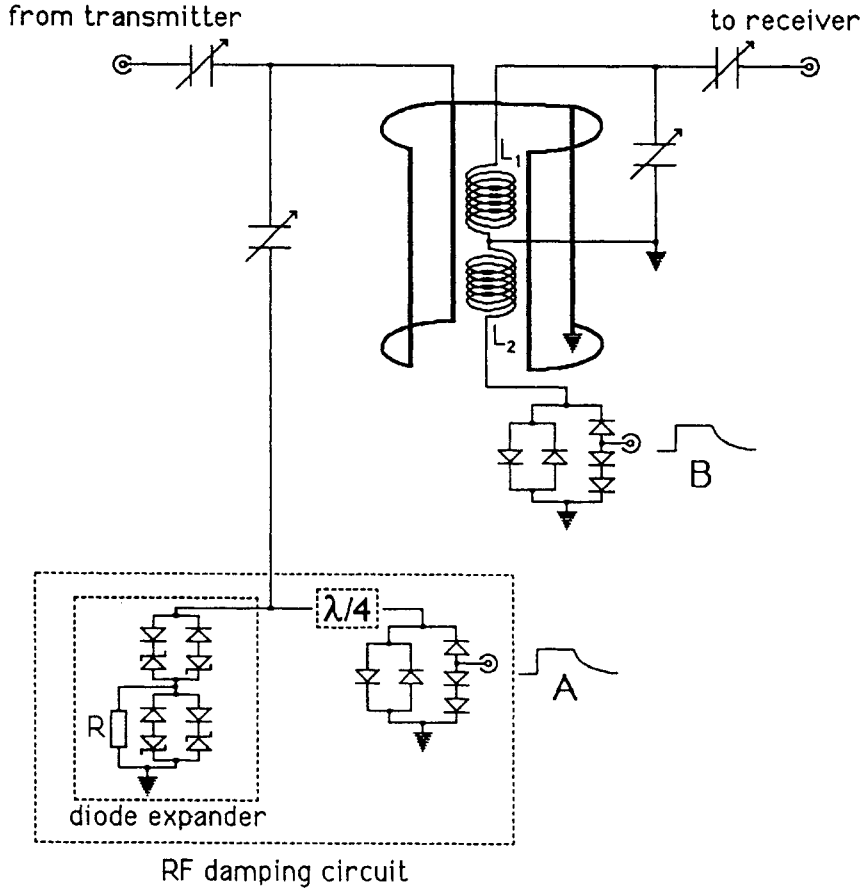


Fig.1. Circuit diagram of the probe. Variable tuning capacitors of 0–100 pF are used. All the silicon diodes are of 1N4148 type; Zener diodes are 1N957B type (6.8 V). The resistor is 1 k Ω , 1/2 W. Pulses A and B can be generated with a damper pulse.

The scheme of the probe is depicted in Fig.1. An ordinary Helmholtz coil is used to transmit the RF power, while the recoil of the nuclear magnetic induction is demanded to a doubly wound coil, whose solenoidal segments (L_1 and L_2) are antiparallely connected. It should be remarked that the isolation reached in this configuration (in the presence of paddles to minimize capacitive couplings [1]) is only 50 dB, a quite low value.

The transmitter coil, which is tuned at 50 Ω and possesses, at 10 MHz, a loaded Q (Q_l) of 44, is connected to a circuitry which is able to efficiently damp the ringing down RF voltage. This circuit, which is an extensive modification of the damping circuit by Andrew and Jurga [13], enables a transitory Q variation of the Helmholtz coil, by means of the suitably tailored pulse A. During the strong RF pulse, the diode expander is in conduction so that

the transmission acts normally towards a circuit tuned at $50\ \Omega$ with $Q_t=44$. At the end of the RF pulse and after the ringing voltage has decreased below the diode expander conduction edge (15 V), the diode expander itself is no more conducting. Now, since the signal A brings the diodes situated on the right of the quarter wave line into conduction, the large impedance seen through such a line becomes important in determining the quick dying out into the noise of the ringing down.

Due to the low isolation between the two coils, a large fraction of the RF pulse is transferred into receiving coil. Here the damping, that in its very nature is an energy dissipation problem, is efficiently performed by the antiparallel arrangement of the two segments in which opposite leakage currents are generated when L_2 is shortcircuited as a result of the pulse B-induced switching of the diodes to their conduction state. At the end of the dissipation, the tail of pulse B opens gradually (to avoid the generation of spikes) the circuit so that L_1 , which is tuned at $50\ \Omega$ to the receiver input and has a $Q(Q_r)$ of 50, becomes able to process the NMR signal.

The temporal relationship between the RF pulse, the A and B pulses and the Q factors of the coils is shown in Fig.2. T shows the voltage pulse sent to the transmitter modulator to generate a RF pulse across the Helmotz coil. From t_1 the pulse rings down reaching about 15 V at t_2 . At this time, the Q_t value, which has remained constant up to t_2 , is abruptly and significantly lowered because of the already described effect of A on the circuit. Indeed, it would have been sufficient to make pulse A starting immediately before t_2 , but for simplicity we trigger it with T. Due to the decreased Q_t value ($Q_t \ll 44$), the signal dies out very quickly into the noise. However Q_t remains in the low value configuration until t_5 , since when it is gradually raised to the initial value, which is reached at the end of the tail of A (t_6).

In the receiving coil, pulse B determines a strong decrease of the Q_r ($Q_r \ll 50$) during the interval $t_0 - t_3$. From t_3 , L_2 is gradually brought up to a configuration of an open circuit, which is reached at t_4 when, on the other hand, the loaded normal Q_r value is reconstituted. Since t_4 on, the receiver coil is ready to process the FID so that the total recovery time is given by $(t_4 - t_1)$.

Figure 3, pattern a, shows the experimental signal that is observed when the probe is matched to a narrow band receiver [19] used to detect our NMR signals. This pattern allowed us to measure a recovery time of 5 μ s; more precisely this time is measured from the end of the RF pulse, t_1 , up to the time t_4 , when we can consider that the signal has died out into the noise. Pattern b of the same figure shows the same signal but in the presence of an NMR sample (plexiglass). Since we are working at the ^1H NMR resonance conditions of the sample (see legend of Fig.3), an FID is superimposed to the pattern a after t_4 . This pattern shows that the S/N ratio obtained with our probe is consistent with the Q_r value [1] and is comparable with that of a signal observed with a single coil probe with a Q of 50 [19].

We believe then that the characteristics of our probe are very good and surely competitive with the other systems discussed in literature. In particular, if an analogous damping procedure is used for a single coil configuration ($Q=50$), as suggested in [13], and the same narrow band receiver is used, a total recovery time of $8\ \mu\text{s}$ is obtained. In other words, the higher damping efficiency of our receiving coil coupled to the lower RF energy to be dissipated are responsible for the faster restoring of the receiver after saturation.

Furthermore the damping of the transmission coil allowed us to use higher values for both Q_t and Q_r with respect to those used by other authors [20] in their crossed coil configuration that, on the other hand, required an higher isolation.

Finally, it should be remarked that the $5\ \mu\text{s}$ recovery time is measured in connection with the use of a narrow band receiver (800 kHz bandwidth) which is able to process minimum transients, indeed, the use of a large band receiver might lead to a further improvement in the dead time, being, in this case, possible a shortening of the damping pulse tails.

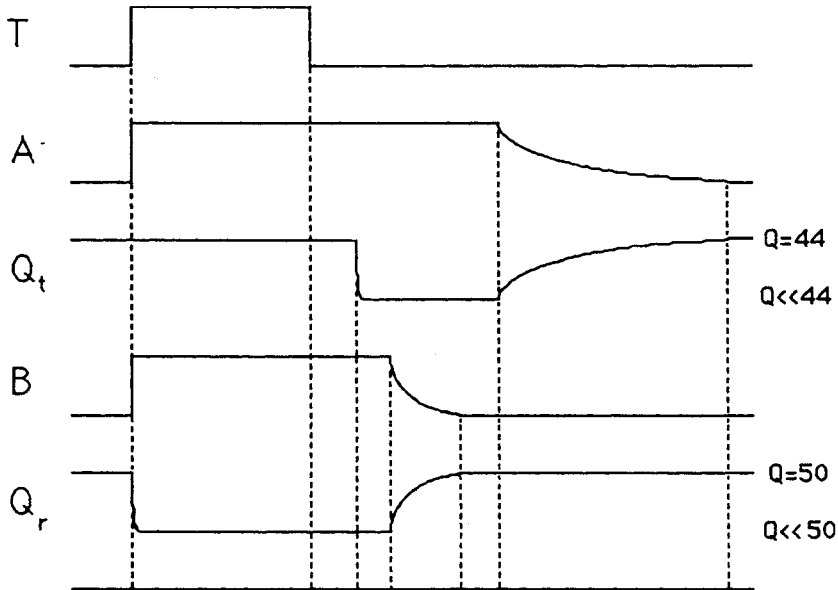


Fig.2. Timing of the probe. The time intervals are not in scale.

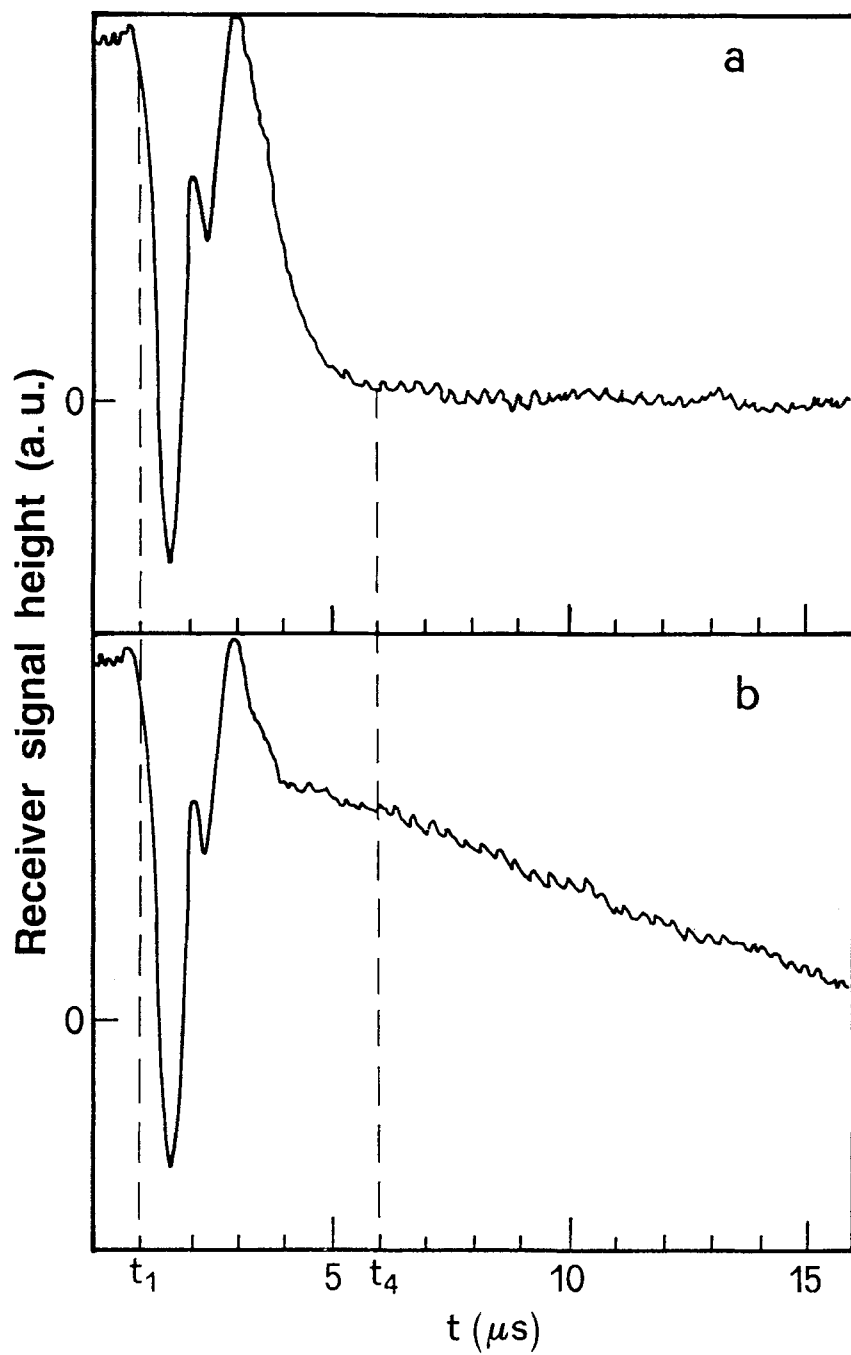


Fig.3. Receiver signals after a RF pulse ($20\ \mu\text{s}$ duration) across our crossed coil probe. Pattern a: empty coil. Pattern b: in presence of a sample of plexiglass at ^1H NMR resonance conditions ($2352.7\ \text{Gs}$, $10\ \text{MHz}$, $t_{\pi/2} = 20\ \mu\text{s}$).

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